

Nicolas Gresset-Bourgeois

PhD candidate · Teaching assistant, engineering school

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Profile

A first-year PhD candidate in computer science specialising in satellite communications, my research focuses on communication mechanisms within nanosatellite swarms, and in particular on routing protocols.

Experience

TeSA
Toulouse, France
October 2025 - Present

PhD candidate in computer science: protocols and reinforcement learning optimisation for nanosatellite swarms

- Study of communication mechanisms and strategies in multi-satellite missions
- Authoring of scientific papers
- Development of a domain-specific simulator in Python 3
- PhD cofunded by CNES (the French space agency) and Viveris

Viveris
Toulouse, France
March 2025 - September 2025

Final-year engineering internship

- Integration of a videoconferencing service into OpenBACH
- In-depth study and analysis of QoE measurement methods for videoconferencing
- Automation of videoconference execution in Python and integration into a framework

Télécom Physique Strasbourg / Innovlab
Strasbourg, France
January 2025 - December 2025

Automated deployment of a UE in a private 5G network

- Year-long engineering project shared between two teams of four students
- Comparative use of several open-source 5G software stacks (OpenAirInterface, srsRAN, Open5GS)
- Technical lead of the project: running progress meetings with the client, report writing, coordination of the teams' work

TU/e
Eindhoven, Netherlands
June 2024 - August 2024

Research assistant internship

- Deployment of a private 5G testbed with OpenAirInterface
- Work spanning a broad range of fields: electronics, software-defined radio (USRPs), virtualisation (Docker)
- Theoretical analysis of the QoS capabilities allowed by the 3GPP specification

Education

Télécom Physique Strasbourg
Strasbourg, France
2022 - 2025

Engineering degree (Diplôme d'ingénieur, MSc level) in computer science and networks

- Coverage of every layer of the TCP/IP stack

Lycée Victor Hugo
Besançon, France
2020 - 2022

CPGE MPSI / MP* (two-year intensive preparatory programme in mathematics and physics)

- Broad scientific grounding and training in mathematical rigour

Skills and interests

Technical skills : C · C++ · Python · Linux: daily user · Git · GitHub · Docker · LaTeX · Typst · NS-3 · P4
Languages : French: native · English: C1 (TOEIC 975/990) · Spanish: B2
Interests : Reading and writing poetry · Sport: rugby, cycling, football · Teaching: private tutoring and adjunct teaching at an engineering school